

Atharv Bhalerao

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EDUCATION

University of British Columbia

Bachelor of Science in Physics

Vancouver, BC

May 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Embedded C, RISC-V Assembly, Verilog/VHDL, Lisp

Microcontrollers & Architectures: PIC33, MSP430, STM32, ARM

Frameworks, Libraries & RTOS: FreeRTOS, OpenMP, OpenCL, AVX (SIMD)

Tools: Linux, Git, Make/CMake, KiCad, LTspice, Docker, Emacs

EXPERIENCE

University of British Columbia

Undergraduate Research Assistant (Dr. Kirk Madison's QDG Lab)

Vancouver, BC

Mar 2026 – Present

- Recipient of EnSURE undergraduate research award.
- Engineered multiple ⁸⁷Rb **Magneto-Optical Traps (MOTs)**, including a 3D vapor-loaded MOT for photo-association and a 2D/3D beam-loaded MOT for a quantum pressure standardization apparatus.
- Executed physical alignment of precision free-space and fiber optics, calculated transition frequencies, and implemented **laser frequency locking** via spectrum analyzers.
- Developed custom **3D Biot-Savart** and beam propagation solvers to determine optimal quadrupole configurations for a 2D MOT and maximize single-mode fiber coupling efficiency.
- Contributed core DAQ architectures to the **QDGbusV1** instrument control system, eliminating legacy technical debt and standardizing experimental automation protocols.

University of British Columbia

Embedded Systems TA, PHYS 319

Vancouver, BC

Jan 2026 – Apr 2026

- Saved an estimated **160+ cumulative hours across a 40-student cohort** by bootstrapping standardized communication protocol libraries (**UART, I2C, SPI**) and automated Makefile flashers, reducing boilerplate setup by **~4 hours/student**.
- Facilitated the **successful deployment of 40+ student** capstone projects by architecting structured debugging methodologies to resolve complex memory, logic and electronic circuit faults in embedded C and assembly **for MSP430 microcontrollers**.

3b Semiconductors pvt ltd

Firmware Developer

Mumbai, IN

Jul 2025 – Aug 2025

- Developed and integrated **real-time dsPIC33 C** firmware modules with a strict 1ms PID control loop, supporting a team-developed dual-stage 1kW LLC power supply.
- Contributed to **microsecond-latency hardware fault protection** (short-circuit, open-loop, OVP, OCP) by deploying an optimized **Binarized Neural Network (BNN) directly on the MCU**, ensuring rapid system isolation prior to component failure.
- Facilitated optimal frequency loop control by leveraging the **BNN's decision speeds** to safely reduce system LR values from **40μH to 32μH**, and contributed to developing a robust **CAN** bus protocol for continuous state telemetry.

PARSEC (UBC)

Electronics and Firmware developer

Vancouver, BC

Nov 2024 – Jun 2025

- Achieved **zero data loss** by engineering **FreeRTOS embedded C firmware on ESP32 Pico** utilizing **mutex-synchronized parallelization for concurrent I2C/SPI sensor DAQ and WiFi telemetry**.
- **Won the CSA's AquaLunar Challenge** by architecting system control PCBs via KiCad for LunaPure, lunar water filtration system.

IIT Bombay

Research Intern

Mumbai, India

Jul 2024 – Aug 2024

- Modeled **IV curves for green GaN MQW microLEDs** by executing **multi-dimensional drift-diffusion models** and constructing equivalent circuit representations in **LTspice**.

PROJECTS

Gesture-Controlled RC Vehicle | *Embedded C, MSP430*

- Engineered a **gesture-based remote-controlled vehicle** using **MSP430 MCU, nRF24L01, and MPU6050 IMU** and wrote firmware for motion interpretation and signal filtering pipeline for smooth control.

LC3 Virtual Machine and Assembler | *C, C++*

- Built a complete **LC3 virtual machine and two-pass assembler** entirely in C, implementing the full instruction set architecture (ISA), **trap routines, memory-mapped I/O**, and precise symbol resolution.

Sphericat: Quantum Mechanics Solver | *C++, OpenGL, FORTRAN*

- Engineered a multi-dimensional Schrödinger equation solver using object-oriented C++ and **finite difference and crank-nicholson** methods to simulate complex quantum mechanical systems.
- Improved memory footprint for large sparse Hamiltonians by implementing **optimized GEMM libraries** and **Arnoldi/Lanczos eigenvalue algorithms**.

High-Performance CNN Framework | *C, AVX, pthreads*

- Engineered a zero-dependency Convolutional Neural Network in C, **accelerating matrix operations by ~8x** via **AVX (SIMD) vectorization, pthreads, and optimized memory caching techniques**.

Instrument Control Library | *Python, GPIB*

- Engineered a **Python/RS232** control library to automate lock-in amplifiers, oscilloscopes, and DMMs, enforcing strict DAQ timing protocols that **reduced measurement uncertainty by 80%** in High- T_c experiments.